STUDENT PERFORMANCE ANALYTICS SYSTEM

Project Documentation

Student	Puja Maharana
Project type	Python data analysis project
Dataset	15 student records in students.csv
Technologies	Python, Pandas, NumPy
Main program	main.py

This report explains the objective, dataset, implementation, key features, results, learning outcomes, and repository location for the Student Performance Analytics System.

1. Project Title

Student Performance Analytics System.

2. Objective

The project reads student marks and attendance from a CSV dataset and calculates totals and averages, assigns grades, determines Pass/Fail results, compares subject performance, identifies top performers, and summarizes attendance. It demonstrates how a dataset can be processed into useful academic summaries.

3. Technologies Used

Python is used for program logic, conditions, loops and functions. Pandas is used to read the CSV, create calculated columns, filter and sort records, and calculate subject averages. NumPy is used to calculate the maximum, minimum and mean of subject marks. The dataset is stored in CSV format.

4. Dataset Description

The file students.csv contains 15 student records. Columns include Student_ID, Name, Python, SQL, HTML, CSS, JavaScript and Attendance. Subject marks are treated as scores out of 100; attendance is represented as a percentage.

5. Implementation

1) Load students.csv with pandas.read_csv(). 2) Select the five subject columns. 3) Calculate Total by summing subject marks and Average by dividing Total by five. 4) Use assign_grade() to assign A for 90+, B for 80–89.99, C for 70–79.99, D for 60–69.99, E for 50–59.99, and F below 50. 5) Use check_resultS() to return Fail if any subject mark is below 40, otherwise Pass. 6) Calculate subject-wise averages, sort by Average to display the top three students, and use NumPy for maximum, minimum and overall mean. 7) Calculate average attendance and list students below 75% attendance.

6. Key Features

• Loads student records from CSV. • Calculates total and average marks. • Assigns grades using a function. • Checks Pass/Fail using the 40-mark subject threshold. • Calculates subject-wise averages. • Displays the top three students. • Uses NumPy to calculate highest mark, lowest mark and mean mark. • Reports average attendance and students below 75%.

7. Output / Results

The program prints the dataset, student count, totals, averages, grades, results, subject-wise averages, top three performers, NumPy statistics, and attendance summary. From the current dataset: 15 student records are processed; the highest individual subject mark is 96; the lowest is 55; the mean of all subject marks is approximately 79.6; the top three students by average are Kavya (93.8), Sneha (93.0), and Neha (91.8); Rohan has 70% attendance and is listed below the 75% threshold.

Screenshots: Two terminal screenshots have been captured to show the program output. Include them with the submission if requested.

8. Final Outcome

The project demonstrates a basic end-to-end workflow for reading structured data, calculating performance metrics, applying grading rules, ranking students and summarizing attendance using Python, Pandas and NumPy.

9. Challenges and Learning

The development process involved organizing CSV data, selecting the subject columns correctly, applying functions row by row, and combining Pandas and NumPy operations. The project provided practice with conditional logic, functions, DataFrame operations, sorting, filtering, arrays and interpreting numerical results.

10. GitHub Repository

Repository URL: https://github.com/pujamaharana029-cpu/Student-Performance-Analytics